vRyan: An AI-Powered Teaching Agent Prototype Exploring Deepfake Question-Answering

Anura Deshpande[1] [0009-0000-5296-1231], Nick Tran[1][0009-0004-9669-2562], Jingyi Huang[1], Naysa Bhargava[1], Ryan S. Baker[2,3], Benjamin D. Nye[1]

[1] Inst. for Creative Technologies, University of Southern California, Los Angeles, CA, USA
[2] Graduate School of Education, University of Pennsylvania, Philadelphia, PA, USA
[3] Centre for Change and Complexity in Learning, University of South Australia, Adelaide, SA 5001, Australia

anurades@usc.edu, nghitran@csu.fullerton.edu, huangyuk@grinnell.edu, nbhargav@usc.edu, rybaker@upenn.edu, nye@ict.usc.edu

Abstract. Virtual Ryan (vRyan) explores the feasibility of an integrated pipeline to produce a deepfake video agent that delivers personalized responses to student questions. Created from online course videos, the system allows learners to interact with a not-at-all-anonymous virtual instructor by submitting queries tied to lecture content. Lecture videos were transcribed and segmented using OpenAI Whisper and embedded for Retrieval-Augmented Generation (RAG). When a question is posed, the RAG model summarizes the fetched content into a coherent answer with an LLM. Next, a synthetic voice modeled from the lecture content is generated using the voice clone. Finally, video frames from the original lectures are combined with the new voice clip to generate the avatar video. vRyan is built entirely with commodity off-the-shelf tools, making it easily replicable (currently: PlayHT for voice clone; SadTalker for video deepfake; and Pinecone with OpenAI ada-002 embeddings for RAG). However, facial rendering remains a bottleneck; while speech responses (of 1-5 sentences) meet real-time thresholds, video response time will be slower and/or require relatively costly GPU cycles. These findings suggest the need for research on a continuum of presence (text, voice, video), to find where video agency is needed.

Keywords: Agent-based learning environments, Adaptive educational systems, LLM-based educational technologies

Demo Link: https://youtu.be/oVIzxT9ghW4

System Presentation

vRyan combines GPT models, RAG, and deepfake tools to create interactive video learning experiences. By enabling students to ask questions, receive personalized video responses, and incorporate a virtual version of their instructor into group discussions, vRyan could increase engagement and proactive exploration of the content. This project has developed a pipeline to explore deepfake designs intended for real-time interaction with students, aiming to explore technologies based on their: 1) Quality: looking and sounding like the human professor; 2) Responsiveness:

replying within a reasonable amount of time and with valid answers; and 3) Replicability: tools that are easily available and could scale up to many classes without being cost-prohibitive. Based on these criteria, a prototype was developed to consider the feasibility of readily-available technologies to support this type of deepfake agent.

vRyan Prototype: To explore these issues, a web-based testbed was developed. First, the learners input a question through the search bar on the vRyan web application, as powered by Node.js. This page also features an introduction video from the avatar as well as a history menu to view previously generated video responses. The primary functionality of interest for this prototype is the ability of the agent to respond and interact with students (i.e., the communication layer for a pedagogical agent). In the current prototype, this functionality has three incremental stages:

1) Text Responses (Retrieval Augmented Generation): Each student input initiates the backend pipeline, which utilizes Pinecone's vector search to locate relevant information from the RAG database of pre-indexed transcripts in order to provide related content [1]. The corpus for the RAG system is based on transcriptions of thirteen videos covering course content on learning analytics, and includes videos that introduce key concepts or facilitate student activities (e.g., reflecting on a discussion). As significant research has already investigated different RAG approaches [2], only major models were explored (e.g., Google vs. OpenAI) and all models were feasible for both quality, responsiveness, and scalability. In the current version, OpenAI models vectorize the text (text-embedding-ada-002) and retrieved content then passes through GPT-4o-mini to generate a tailored and relevant response to the user [3]. A text preview of the response is shown to the user while the video is being generated.

2) Speech Cloning: The text is converted into a speech response intended to sound like the professor. A list of popular tools was compiled, and after evaluating four we identified at least one feasible choice (PlayHT API [4]). In this API, the synthetic voice is a modeled voice clone from the audio of the instructor's lecture content, created online in the PlayHT studio prior to real-time generation. Overall, this voice sounded recognizably similar to the original, responded fast enough for real-time dialog, and costs were comparable to other synthetic voices. Other hosted services were also feasible, but we did not test open-source models of similar quality.

3) Video Deepfake: Tools for video generation showed greater tradeoffs between quality, responsiveness, and cost. Opting to examine the quality of a lower-cost option, we implemented SadTalker, a self-hosted open-source tool [5]. This produces a lip-synced deepfake of the instructor delivering the response, based on a combination of the speech-clone voice file and also an original lecture video frame. The final video is then encoded through FFmpeg and presented to the learner on the web application page. Overall, SadTalker videos ranged from reasonably good quality (with slower rendering settings) or acceptable quality (with faster settings). However, in both cases responses were far too slow for real-time use (e.g., 5m to 40m rendering per answer). On the converse, hosted APIs were typically $1 - $3 per minute of video, likely infeasible for this use-case, even if responses were fast enough.

Contribution and Advantages. Pre-recorded videos and static text-based responses can feel inflexible and impersonal. vRyan found that it was feasible to generate specific and engaging video responses through a unified pipeline of ML tools, particularly for quality (semi-realistic for 1-5 sentence answers) and replication (relies on well-maintained tools). This research confirms that research on pedagogical agents may increasingly leverage video deepfakes of teachers, particularly when responses are not real-time or if only audio answers are required.

Limitations. Real-time generation of video deepfakes remains challenging, and our exploration suggests significant sacrifices for response time, quality, and/or computational cost. While systems like VASA-1 demonstrate real-time capabilities, these are not yet widely available and require a substantial GPU (e.g., nVidia RTX 4090) [6]. As such, generation time remains a significant bottleneck. Video quality is also constrained by compute limits, with even limited improvements to lip synchronization increasing processing time. Finally, current tools typically only animate the face, but do not add non-verbal motions such as posture or hand gestures (which professors commonly use). Ethical issues for deepfakes must also be considered, such as how it changes perceptions of the professor or other support.

Discussion

In the context of online learning, vRyan shows a path forward to scale up instructor presence, where the same virtual instructor might at times respond using text, audio, or video. Further research should explore when more-costly modes (e.g., video) offer the most impact, such as for personalized help or increasing participation in student discussions. In addition, the combination of audio, video, and text interfaces accommodates diverse learning preferences.

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References

1. Pinecone: The Vector Database to Build Knowledgeable AI. https://www.pinecone.io/, last accessed 2025/05/06
2. Swacha, J., Gracel, M.: Retrieval-Augmented Generation (RAG) Chatbots for Education: A Survey of Applications. *Appl. Sci.* **15**, 4234 (2025). https://doi.org/10.3390/app15084234
3. OpenAI: API Platform. https://openai.com/api/, last accessed 2025/05/06
4. PlayHT: #1 Free AI Voice Generator, Text to Speech, & AI Voice Over. https://play.ht/, last accessed 2025/05/06
5. SadTalker: Learning Realistic 3D Motion Coefficients for Stylized Audio-Driven Single Image Talking Face Animation. https://github.com/OpenTalker/SadTalker, last accessed 2025/05/06
6. Xu, S., Chen, G., Guo, Y.-X., Yang, J., Li, C., Zang, Z., Zhang, Y., Tong, X., Guo, B.: VASA-1: Lifelike Audio-Driven Talking Faces Generated in Real Time. In: *Advances in Neural Information Processing Systems*, vol. 37, pp. 660–684 (2024)